MicroWorld Third-Party Library Build Guide

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1. Overview

This document describes where the source code of all third-party libraries is stored, which versions are used, which `.so` files they produce, and how to build the key components (fontconfig, poppler, OpenSSL, libmagic).

Item	Value
Source code path	`/tmpdata/omkar_new_updated`
Install prefix (final libs)	`/opt/MicroWorld`
Local prefix (fontconfig)	`/tmpdata/omkar_new_updated/local`
Debian build machine	`192.168.2.216` (host: `ketan-VirtualBox`)
RPM build machine	`192.168.2.218` (same paths)
RPM build commands	`compile.txt` in `/tmpdata/omkar_new_updated`

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2. Library Versions and Output Files

Library	Version	Output `.so` file
libpng	1.6.54	`libpng16.so.16`
libtiff	4.7.1	`libtiff.so.6`
libjpeg-turbo	3.1.3	`libjpeg.so.62`
libopenjp2 (OpenJPEG)	2.5.4	`libopenjp2.so.7`
libarchive	3.8.5	`libarchive.so.13`
Leptonica	1.87.0	`libleptonica.so.6`
FreeType	2.14.1	`libfreetype.so.6`
Tesseract	5.5.2	`libtesseract.so.5`
Poppler	23.12.0	`libpoppler.so.133`
curl	8.12.0	`libcurl.so.4`
PCRE2	10.42	`libpcre2-8.so.0`
file (libmagic)	–	`libmagic.so.1` and `magic.mgc` (magic database)

Additional sources present in the directory: autoconf-2.71, expat-2.6.2 / 2.7.4, zlib-develop, cups-2.4.16, libcups, fontconfig-2.17.1, libharu, openssl-3.4.0.

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3. Directory Layout (`/tmpdata/omkar_new_updated`)

Directory / File	Purpose
`libjpeg-turbo`, `lpng1654`, `tiff-4.7.1`, `openjpeg-master`, `libarchive-master`, `leptonica-master`, `freetype`, `tesseract-main`	Image / document library sources
`fontconfig-2.17.1`	Fontconfig source (dependency of poppler)
`poppler_new`	Poppler 23.12.0 source and build directory
`file`	Source of `file`, used to generate `magic.mgc` and `libmagic.so.1`
`openssl-3.4.0`, `openssl_compile.txt`	OpenSSL source and its compile steps
`expat-*`, `zlib-develop`, `cups-2.4.16`, `libharu`	Other dependency sources
`local`	Local install prefix for fontconfig
`libs`, `local-pc`, `tools`, `link.txt`	Supporting files
`contentControl.zip` / `contentControl`	Runtime content-control package (see section 7)

Related paths outside this directory:

- `/tmpdata/curl-8.12.0/` – curl source, produces `libcurl.so.4`
- `/tmpdata/cmake-3.28.1/bin/cmake` – use this CMake if the system CMake is too old

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4. Building Poppler 23.12.0

4.1 Compiler setup

```
export CC=/usr/bin/gcc-8
export CXX=/usr/bin/g++-8
```

4.2 Build and install fontconfig (prerequisite)

```
cd /tmpdata/omkar_new_updated/fontconfig-2.17.1
./configure --prefix=/tmpdata/omkar_new_updated/local
make
make install
```

4.3 Environment variables

```
export PKG_CONFIG_PATH=/tmpdata/omkar_new_updated/local/lib/pkgconfig:$PKG_CONFIG_PATH
export LD_LIBRARY_PATH=/opt/MicroWorld/lib:$LD_LIBRARY_PATH
export LDFLAGS="-L/opt/MicroWorld/lib -Wl,-rpath,/opt/MicroWorld/lib"

# verify fontconfig is found
pkg-config --modversion fontconfig
```

4.4 Configure with CMake

The machine has an old CMake, so use CMake 3.28.1. Run this from the build directory:

`/tmpdata/omkar_new_updated/poppler_new/poppler-23.12.0/build` (the directory that sits next to `CMakeLists.txt`, hence `..`)

```
/tmpdata/cmake-3.28.1/bin/cmake .. \
  -DCMAKE_INSTALL_PREFIX=/opt/MicroWorld \
  -DENABLE_NSS3=OFF \
  -DENABLE_GPGME=OFF \
  -DTIFF_LIBRARY=/opt/MicroWorld/lib/libtiff.so \
  -DTIFF_INCLUDE_DIR=/opt/MicroWorld/include \
  -DENABLE_QT5=OFF \
  -DENABLE_QT6=OFF \
  -DENABLE_BOOST=OFF \
  -DENABLE_LIBCURL=ON \
  -DCURL_LIBRARY=/tmpdata/curl-8.12.0/lib/.libs/libcurl.so \
  -DCURL_INCLUDE_DIR=/tmpdata/curl-8.12.0/include
```

4.5 Fix rpath for `libpoppler-cpp.so`

Edit this file:

```
/tmpdata/omkar_new_updated/poppler_new/poppler-23.12.0/build/cpp/CMakeFiles/poppler-cpp.d
```

Add the following at the **end** of the line:

```
-Wl,-rpath,/opt/MicroWorld/lib: /opt/MicroWorld/lib/libpoppler.so.133
```

Then run `make` and `make install`.

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5. Building OpenSSL 3.4.0

Steps are saved in `/tmpdata/omkar_new_updated/openssl_compile.txt`:

```
cd /tmpdata/omkar_new_updated/openssl-3.4.0
make clean 2>/dev/null

./Configure linux-x86_64 \
    --prefix=/opt/MicroWorld \
    --openssldir=/opt/MicroWorld/ssl \
    --libdir=lib \
```

```
    shared \
    -Wl,-rpath=/opt/MicroWorld/lib \
    -Wl,--enable-new-dtags \
    -L/opt/MicroWorld/lib

make -j$(nproc)
make install_sw
```

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6. libmagic and magic.mgc

The `file` source directory (`/tmpdata/omkar_new_updated/file`) is used to generate:

- `magic.mgc` – the compiled magic database
- `libmagic.so.1` – the libmagic shared library

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7. contentControl Package

`contentControl.zip` is stored in the same path (`/tmpdata/omkar_new_updated`). The extracted `contentControl` directory contains the tools and data files used at runtime:

Item	Type	Size	Date	Description
`tessdata`	Directory	–	Oct 24 2025	Tesseract OCR language data
`pdfimages`	Executable	70K	Oct 24 2025	Poppler utility that extracts images from PDF files
`antiword_DATA`	Directory	–	Oct 24 2025	Data files used by antiword
`antiword`	Executable	223K	Oct 24 2025	Extracts text from legacy Word (.doc) files
`magic.mgc`	File	11M	Sep 9	Compiled magic database (generated from the `file` source, see section 6)

Notes:

- `magic.mgc` here is the output of the `file` build described in section 6.
- `pdfimages` comes from the poppler build in section 4 and relies on the libraries in `/opt/MicroWorld/lib` .
- `tessdata` is used by Tesseract 5.5.2 (`libtesseract.so.5`).

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8. Notes

- All libraries install into `/opt/MicroWorld` with rpath set to `/opt/MicroWorld/lib` , so they load without changing system libraries.
- The same directory layout is used on both machines. Debian build: `192.168.2.216` . RPM build:

`192.168.2.218`, with build commands in `compile.txt`.

- Poppler build depends on libtiff and libcurl being built first, and on fontconfig from the local prefix.
- Recommended build order: zlib, expat, libpng, libjpeg-turbo, libtiff, openjpeg, freetype, fontconfig, curl, OpenSSL, libarchive, leptonica, tesseract, poppler, file/libmagic. (Order is a suggestion based on dependencies; adjust to your existing process.)